XIANYU CHEN

Kenneth H. Keller Hall 6-240, 200 Union St SE, Minneapolis, MN, 55455
(612)-814-3793 ⋄ chen6582@umn.edu ⋄ chenxy99.github.io ⋄ Google Scholar

EDUCATION

University of Minnesota, Twin Cities (GPA: 4.0/4.0) Sept. 2019 - Now
Ph.D. in Computer Science (anticipated to graduate in 06/2024)
Advisor: Dr. Catherine Qi Zhao

Sun Yat-sen University Sept. 2015 - June 2018
M.E. in Information and Communication Engineering
Advisor: Dr. Ming Jiang

Sun Yat-sen University Sept. 2011 - June 2015
B.E. in Communication Engineering
Advisor: Dr. Ming Jiang

RESEARCH INTEREST

Computer Vision, Machine Learning, Deep Learning, Nature Language Processing

PUBLICATION

Journals

- **Xianyu Chen**, Yali Wang, Jianzhuang Liu and Yu Qiao. DID: Disentangling-Imprinting-Distilling for Continuous Low-Shot Detection. *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 7765-7778, July 2020.
- **Xianyu Chen** and Ming Jiang. Enhanced Adaptive Polar-Linear Interpolation Aided Channel Estimation. *IEEE Wireless Communications Letters (WCL)*, vol. 8, no. 3, pp. 693-696, June 2019.
- Huaiyin Lu, Lin Zhang, **Xianyu Chen** and Zhiqiang Wu. Recursive Carrier Interferometry Aided High Data Rate OFDM Systems with PAPR Suppression, Phase Noise Rejection and Carrier Frequency Offsets Compensation. *IEEE Transactions on Vehicular Technology (TVT)*, vol. 68, no. 4, pp. 3655 - 3671, April 2019.
- Kuan Wu, Ming Jiang, Fengxia She and **Xianyu Chen**. Relay-Aided Request-Aware Distributed Packet Caching for Device-to-Device Communication. *IEEE Wireless Communications Letters (WCL)*, vol. 8, no. 2, pp. 217-220, Feb. 2019.
- Zhengpeng Li, Ming Jiang, Xiaona Zhang, **Xianyu Chen** and Weikun Hou. Space-Time-Multiplexed Multi-Image Visible Light Positioning System Exploiting Pseudo-Miller-Coding for Smart Phones. *IEEE Transactions on Wireless Communications (TWC)*, vol. 16, no. 12, pp. 8261-8274, Dec. 2017.
- **Xianyu Chen** and Ming Jiang. Adaptive Statistical Bayesian MMSE Channel Estimation for Visible Light Communication. *IEEE Transactions on Signal Processing (TSP)*, vol. 65, no. 5, pp. 1287-1299, March 2017.

Conferences

- Jinhui Yang*, **Xianyu Chen***, Ming Jiang, Shi Chen, Louis Wang and Qi Zhao. VisualHow: Multimodal Problem Solving. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. (*Co-first authors/Equal contribution)
- **Xianyu Chen**, Ming Jiang and Qi Zhao. Leveraging Human Attention in Novel Object Captioning. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.

- **Xianyu Chen**, Ming Jiang and Qi Zhao. Predicting Human Scanpaths in Visual Question Answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- **Xianyu Chen**, Ming Jiang and Qi Zhao. Self-Distillation for Few-Shot Image Captioning. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021.
- Ze Yang, Yali Wang, **Xianyu Chen**, Jianzhuang Liu and Yu Qiao. Context-Transformer: Tackling Object Confusion for Few-Shot Detection. In *AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
- **Xianyu Chen** and Ming Jiang. Low-Complexity Adaptive Channel Estimation. In *Proceedings of the Vehicular Technology Conference (VTC)*, 2018.
- Yufa Chen, Ming Jiang, Lin Zhang and **Xianyu Chen**. Polarity Modulated Complex Colour Shift Keying for OFDM-Based Visible Light Communication. In *Proceedings of the IEEE/CIC International Conference on Communications in China (ICCC)*, 2017.
- Zhengpeng Li, Ming Jiang, Xiaona Zhang, **Xianyu Chen** and Weikun Hou. Miller-Coded Asynchronous Visible Light Positioning System for Smart Phones. In *Proceedings of the IEEE Vehicular Technology Conference (VTC)*, 2017.
- **Xianyu Chen** and Ming Jiang. Enhanced Bayesian MMSE Channel Estimation for Visible Light Communication. In *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2016.

TALK

- "Leveraging Human Attention in Novel Object Captioning", presented at *International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.
- "Self-Distillation for Few-Shot Image Captioning", presented at *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021.

SERVICE

Review for Journals

- IEEE Transactions on Neural Networks and Learning Systems — 2023-Now
- Journal of Visual Communication and Image Representation — 2021-Now

Review for Conferences

- International Conference on Computer Vision (ICCV) — 2023
- European Conference on Computer Vision (ECCV) — 2022
- Computer Vision and Pattern Recognition (CVPR) — 2022, 2023
- Winter Conference on Applications of Computer Vision (WACV) — 2022, 2023, 2024
- AAAI Conference on Artificial Intelligence (AAAI) — 2023

EXPERIENCE

Visual Information Processing Lab, University of Minnesota — Sept. 2019 - Now
Computer Vision, Vision and Language, Machine Learning

- Proposed a large-scale dataset enabling a family of new vision-language tasks and computational methods for understanding and solving real-life problems.

- Designed a new method to complement novel object captioners with human attention features characterizing generally important information independent of tasks.
- Designed a new deep reinforcement learning method to predict scanpaths leading to different performances in visual question answering.
- Developed an ensemble-based self-distillation method for image captioning with few paired data and a large number of unpaired images and captions.
- Designed a new method for few-shot object detection by leveraging bottom-up and top-down attention based on object-concentration loss and background-concentration loss.

Multimedia Research Center, Shenzhen Institutes of Advanced Technology Chinese Academy of Sciences Aug. 2018 - Dec. 2018
Computer Vision in Low-Shot Object Detection

- Designed a simple but effective solution for continuous low-shot detection based on architecture design (Disentangling), model initialization (Imprinting), and training methodology (Distilling).

Wireless Communication Lab, Sun Yat-sen University Dec. 2013 - June 2018
Estimation Theory in Optical Wireless Communication

- Designed a dimming control framework achieving optimal waveform design and extended it into two different variants to acquire better performance.
- Designed an enhanced adaptive polar linear interpolation to estimate the channel information more robustly and efficiently.
- Designed a new channel estimation method and analyze its performance bound based on Bayesian MMSE estimation.

TEACHING

Teaching Assistant

- CSCI 2033: Linear Algebra — Spring 2023
- CSCI 5521: Machine Learning Fundamentals — Fall 2022
- CSCI 5302: Analysis of Numerical Algorithms — Spring 2022
- CSCI 5521: Machine Learning Fundamentals — Fall 2021

HONOR

- College Science and Engineering Graduate Fellowships — University of Minnesota, 2019, 2020
- National Scholarship — Department of Education of China, 2013, 2017
- Outstanding Undergraduate Student Award — Sun Yat-sen University, 2015
- Zhentai Scholarship — Sun Yat-sen University, 2012

SKILL

Programming Language: Python, Matlab, JavaScript, HTML, C/C++, Linux shell

Tools: Pytorch, Tensorflow, Keras, Opencv, Unix/Linux, Git, Scikit-Learn, LaTeX